

Duality is a structure; the sector is the cause

T-duality of the compact boson stated on its sectors and checked by two kernels, the measured invariant it predicts, and a criterion for when a self-dual point is physical

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SocrateAI-Scientific-QuantumFluids

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Abstract

An involution $R \leftrightarrow \alpha'/R$ on a length scale was, in earlier work of this programme, offered a physical home in T-duality, in K3 mirror symmetry, and in the modular symmetry of the quantum Hall system. Three rounds of pre-registered numerics and twenty-eight machine-checked theorems later, we can say where such a duality lives and what it does. We state T-duality on the sectors $(n, w) \in \mathbb{Z}^2$ of the compact boson and prove, in Lean 4 against Mathlib with two independent kernels, that (i) it is a reindexing of the sector lattice that leaves the sector sum invariant with no convergence hypothesis, (ii) the product of the electric and magnetic scaling dimensions is radius-independent, $\Delta_e \Delta_m = \frac{1}{4}$, which in the superfluid dictionary is $\eta n_s \lambda_T^2 = 1$ – the relation this programme measured to 12–27% on eleven energies in two boxes – and (iii) the self-dual radius $\sqrt{2}$ and the radius $2\sqrt{2}$ at which the vortex becomes marginal are different points: the Berezinskii–Kosterlitz–Thouless transition is the threshold of one sector, not the fixed point of the exchange. From these three facts and the literature on Ising, string-gas and superconductor–insulator dualities we extract a criterion: a self-dual point pins a transition only when the dual sectors are simultaneously critical on a discrete set of fixed points; on a line of fixed points it is a point like any other; when both sectors are populated it can be a preferred equilibrium. In every case what acts is a sector – a conserved label that crosses a threshold, proliferates, annihilates, or is imposed – and the duality relabels it. The duality is exact, verified, and not a cause. No physics here is new; the mathematics is elementary; what is new is that the distinction is stated so that a kernel can check it and an experiment can test it.

1 Three questions one involution raised

The dual-length involution $k \mapsto k_s^2/k$ of [1] has the form of T-duality. It was pointed out that the same involution is the Fricke element of $\Gamma_0(N)^+$, the mirror involution of degree-2N lattice-polarised K3 surfaces [2], and that $\Gamma_0(2)$ acts on the quantum Hall conductivity plane [3]. Each identification was taken to its gate [4, 5]: the Fricke matrix normalises Mathlib’s $\Gamma_0(n)$ and its axis action is the dual-length map, but no $\Gamma_0(n)$ acts on a wavenumber (`Fricke.lean`); the Fricke element fixes the Hall critical point’s orbit but sends every plateau off the odd-denominator class (`QHFricke.lean`); and the physical BKT point is not the self-dual radius. The residue of these three verdicts is one question: if a duality is exact and verified, and yet is not where the physics happens, what is it, and what is?

2 T-duality on the sector lattice, machine-checked

Let the compact boson have radius R in the normalisation where the self-dual radius is $\sqrt{2}$. The primary of momentum n and winding w has $h = \frac{1}{2}(n/R + wR/2)^2$, $\bar{h} = \frac{1}{2}(n/R - wR/2)^2$, hence

$$\Delta(n, w; R) = n^2/R^2 + w^2R^2/4, \quad s = h - \bar{h} = nw.$$

The electric (spin-wave, $e^{i\theta}$) operator is $(1, 0)$; the magnetic (vortex) operator is $(0, 1)$. The following are theorems of `CompactBoson.lean` (twelve statements, axiom footprint `{propext, Classical.choice, Quot.sound}` accepted by the Lean kernel and by `nanoda` through `Comparator`; three negative controls – product $\frac{1}{2}$, BKT at $\sqrt{2}$, duality with $R \mapsto 1/R$ – fail).

Theorem 1 (duality as reindexing). $\Delta(n, w; R) = \Delta(w, n; 2/R)$ and $s(n, w) = s(w, n)$. Consequently, for every weight F , $\sum_{(n,w) \in \mathbb{Z}^2} F(\Delta(n, w; R), s) = \sum_{(n,w) \in \mathbb{Z}^2} F(\Delta(n, w; 2/R), s)$.

The second statement is `Equiv.tsum_eq` applied to the swap of $\mathbb{Z}^2$: reindexing a sum by a bijection is an identity of `tsum` whether or not the series converges. This is what the duality is – a bijection of the sector lattice that preserves the spectrum. Savit’s review [6] gives the general form: a duality is a Fourier transform on the group of sectors, electric charges in G against magnetic charges in $\hat{G}$, with the Lagrangian sum over one family turned into the Hamiltonian sum over the dual family by Poisson summation.

Theorem 2 (the invariant). $\Delta(1, 0; R) \Delta(0, 1; R) = \frac{1}{4}$ for every $R \neq 0$.

In the XY / two-dimensional-superfluid dictionary $\Delta(1, 0) = \eta/2$ and $\Delta(0, 1) = n_s \lambda_T^2/2$, so the invariant reads $\eta n_s \lambda_T^2 = 1$. That is the relation measured in [5]: 1.047 and 1.072 on two equilibrated energies in round 2, 1.12–1.22 on the six energies of the heating ladder, 1.14–1.27 on the five energies of the $L = 32$ box – with the finite-size excess predicted by Hasenbusch’s logarithmic corrections [7]. *The numerical verification of the duality in this programme is the measurement of this product.*

Theorem 3 (self-dual is not marginal). For $R > 0$: $\Delta(1, 0) = \Delta(0, 1)$ iff $R = \sqrt{2}$, where both equal $\frac{1}{2}$; $\Delta(0, 1) = 2$ iff $R = 2\sqrt{2}$, where $\Delta(1, 0) = \frac{1}{8}$, i.e. $\eta = \frac{1}{4}$; and $2\sqrt{2} \neq \sqrt{2}$.

The transition is where the vortex operator becomes marginal – Nelson and Kosterlitz’s universal jump $n_s \lambda_T^2 = 4$ [8], located in our own runs at $T_{\text{BKT}} = 0.821$ with $\eta = 0.31$ there – and that is a condition on one sector. The exchange symmetry has its fixed point elsewhere, and nothing happens there.

3 When is a self-dual point physical?

The compact boson is the case with a *line* of fixed points, one for each R . Three other cases complete the criterion.

Ising: yes. Kramers–Wannier self-duality [9] fixes T_c at $\sinh 2K = 1$ because the order and disorder operators [10] exchange under the duality and $\mathbb{Z}_2$ has a single symmetric point: the dual sectors are simultaneously critical and the fixed-point structure is discrete.

String gas cosmology: a preferred equilibrium. In the Brandenberger–Vafa scenario [11, 12] “the symmetry we make use of is T-duality, and the new degrees of freedom are the string oscillatory modes and the string winding modes”. Momentum modes, with energy n/R , resist contraction; winding modes, with energy wR , resist expansion; when both are populated the radion is stabilised at the self-dual radius as the minimum of an effective potential. The self-dual point is physical there because both dual sectors are present and compete. And the exit from the Hagedorn phase – the event that makes three dimensions large – is the *annihilation of winding modes*: a sector event, with T-duality as the symmetry of the spectrum in which it is described.

Superconductor–insulator transition: a constraint. Boson–vortex duality [13, 14] predicts, if the transition is self-dual, a universal resistance $h/4e^2$ [15, 16]; measured values scatter around it. The cause of the transition is the proliferation of vortices on one side and of charges on the other; self-duality constrains the critical point only if the fixed point happens to be self-dual, which is a property of the fixed point, not of the duality.

Quantum Hall: organisation, not cause. $\Gamma_0(2)$ organises the plateau diagram and the position of the critical points [3]; the cause of the quantisation is Laughlin’s gauge argument [17] on the Chern sector; and the Fricke element, which fixes the critical orbit, is not a symmetry of the plateaux (`QHFricke.lean`).

Criterion. A self-dual point pins a transition iff the two dual sectors are simultaneously critical and the fixed-point set is discrete. On a line of fixed points the self-dual point is a point like any other and the transition sits where one sector crosses its own threshold. When both dual sectors are populated and compete energetically, the self-dual point is a preferred equilibrium. In each case the duality relabels the sectors and the sector acts.

4 What acts: the sector

The same triad recurs in every case above: a group of sectors ($\mathbb{Z}$ for $U(1)$ windings, $\mathbb{Z}_2$ for Ising, Chern numbers, string windings); a duality that is a Fourier or reindexing map between G -sectors and $\hat{G}$ -sectors preserving the spectrum; and physics that happens when a sector crosses a threshold, proliferates, annihilates, or is imposed. The causal role of the sector is what the rest of this programme established [5]: it is conserved by every continuous evolution without a phase slip (`TopologicalProtection`), additive in scale so that a bound pair is invisible from outside (`ScaleResolvedWinding`), read correctly by the sampled loop sum as the continuum degree (`ContinuumWinding`), it carries its own temperature (`SectorTemperature`), and in an energy-matched intervention it is a difference-maker: the same energy delivered as topology lowered the condensate by 0.49–0.73 where phonons lowered it by at most 0.035, with the injected sector reading five to eight times the bath’s temperature on our own calibrated thermometer. That is why one involution appears in helium, Hall bars, K3 moduli and string gases, and why it explains none of them.

5 What this paper does not claim

- Novelty in the mathematics: the three theorems are elementary; their value is that the distinction between the duality and the sector is stated so that a kernel checks it.
- Novelty in the physics: Kramers–Wannier, Nelson–Kosterlitz, Brandenberger–Vafa, Fisher and Lütken–Ross are cited, not extended.
- Any identification of a superfluid with a K3 surface or a string background. The K3 mirror involution is the same map on a different space; the criterion above is what the shared map does and does not license.
- That every duality in physics fits the triad: the cases examined do; the general statement is a reading.

Addendum (2026-09-26): the criterion sharpened by the non-invertible-symmetry literature

A literature review after publication located the modern machinery that makes §3’s criterion precise rather than descriptive. Aasen, Mong and Fendley state the core distinction directly: a self-duality is implemented by an operator that “is not a symmetry in the traditional sense... not unitary or even invertible” [18]. Choi, Córdova, Hsin, Lam and Shao exhibit the Poisson resummation that turns this into an explicit lattice statement, and prove that at some self-dual

points a mixed 't Hooft-like anomaly *forbids* a trivially gapped phase – an anomaly argument, not the duality map, is what does the forbidding [19]. Kaidi, Ohmori and Zheng give the clean mechanism for our own already-published observation: at $SO(3)$ Yang–Mills $\theta = \pm\pi$, the self-dual point sits *between two physically inequivalent gapped phases* (a $\mathbb{Z}_2$ TQFT and the trivial phase), which is why it is forced to be a transition [20]. Shao’s lecture notes give the general rule, in one sentence: away from a critical point Kramers–Wannier duality “is not a duality relating two equivalent descriptions of a single system... it is a map from the high-temperature phase to the low-temperature phase,” and at the compact boson’s self-dual radius $\tau = i$ “the duality defect reduces to an invertible $\mathbb{Z}_4$ symmetry” [21] – exactly why $R = \sqrt{2}$ (a point on a continuous moduli space with enhanced symmetry) is structurally unlike the Ising or $SO(3)$ -Yang–Mills self-dual points (between two gapped, inequivalent phases), and why neither is like the vortex-marginality radius $2\sqrt{2}$, which is a threshold on one sector alone. One further correction: Hayashi and Tanizaki show that the Cardy–Rabinovici $SL(2, \mathbb{Z})$ “self-duality” of the 2D Coulomb gas/ $\mathbb{Z}_N$ clock model is not naive – it holds only after gauging the electric one-form symmetry with a discrete θ -term – and that the resulting self-dual point, where three first-order transition lines meet, is again forced non-trivial by a separate mixed gravitational anomaly, not by the duality symmetry itself [22]. None of this changes a theorem of this paper; it replaces the numerical observation “ $\sqrt{2} \neq 2\sqrt{2}$ ” with an explicit criterion (inequivalent sectors on the two sides $\Rightarrow$ forced transition; a continuous moduli space with enhanced symmetry $\Rightarrow$ no transition) that a future module could formalise as a finite-group gauging statement.

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